

Jingyi Zhang

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Research Interests

My research focuses on human-computer interaction, virtual characters and agents, and asymmetric collaboration across augmented and virtual reality. I explore creative applications and novel interaction paradigms for immersive systems in performance, cultural heritage, sports, and mental health. My interests also include the integration of artificial intelligence into virtual and augmented reality.

Education

PhD Computer Science | University College London

04/2023 - 05/2027 (expected)

Supervisors: Prof. Anthony Steed (Deputy Head, Department of Computer Science), Prof. Ifat Yasin (Vice-Dean - Equality, Diversity and Inclusion, Faculty of Engineering)

MSc Computer Graphics, Vision and Imaging | University College London Distinction

09/2021 - 09/2022

Modules: Machine Learning for Visual Computing | Image Processing | Computer Graphics | Machine Vision | Numerical Optimisation | Perception and Interfaces | Robot Vision and Navigation | Virtual Environment

Thesis: [“Supporting Co-Presence in Virtual Environments by Actor Control of Multiple Avatars”](#), **Supervisor:** Prof. Anthony Steed
Developed to a [Paper submitted to ISMAR 2023](#)

BSc Computer Science | University College London 1st Class Honours

09/2018 - 06/2021

Thesis: [“Removal of Artefacts in Digitised Medical Optical Microscopy”](#), **Supervisor:** Prof. Delmiro Fernandez-Reyes

Work Experience

Teaching Assistant | University College London

09/2023 - now

- COMP0016 Systems Engineering (Undergraduate): Guided student teams in project management and provided Q&A support; coursework marking.
- COMP0113 Virtual Environments (Master’s): Delivered tutorials on Unity, C# programming, deploying VR applications to headsets, and networked VR development with Ubiq; lab Q&A; seminar and coursework marking.
- COMP0026 Image Processing (Master’s): Lab Q&A and coursework marking.
- COMP0122 MSc Computer Graphics, Vision and Imaging Project: Assisted faculty supervisors with the supervision of MSc dissertation projects.

Research Assistant | University College London

Nov 2022 - Dec 2022

- Conducted Research on comparing agent representation in mixed reality. **Developed to a [Paper submitted to VRST 2023](#).**

Student Summer Internship | Tencent (Shenzhen, Guangdong, China)

Aug 2021 - Sep 2021

- Collaborated with senior researchers to produce demo of Volumetric video and motion capture technique.
- Responsible for writing group project plan and progress report.
- Wrote speech scripts and PPT for technology conference.
- Wrote an industry survey about metaverse techniques and companies.

Student Summer Internship | Microsoft (Guangzhou, Guangdong, China)

Jul 2020 - Aug 2020

- Developed a speech to speech translation software for conference use by calling Microsoft Azure Cognitive Service API, and released the tutorial in official account to promote it and the Neural Speech Service. [Wechat Article](#)
- Used MXchip and Raspberry Pi as the physical device, popularized the scientific applications of the Internet of Things, Microsoft Azure, Power BI through small projects such as Auto Voice Weather Forecast and Word Cloud, which was published on country's biggest video-sharing site. [Bilibili Video](#)
- Wrote a tutorial about how to train a reinforcement learning agent with Azure Machine learning and its future applications.

Student Summer Internship | Intel (Shanghai, China)

Jun 2019 - Aug 2019

- Conducted research on the AI ecosystem and contributed to industry reports.
- Project matchmaking with AI startup companies.

Publications

 Head-Mounted Augmented Reality Enhances Affective and Perceptual Responses to Vigorous Intermittent Lifestyle Physical Activity	Under review
<i>Journal of Sport and Health Science</i>	
Jonathan M. Bird, Henry W. W. Potts, Harold Kwong, Jingyi Zhang , John J. Mitchell, Aidan Acquah, Joanna M. Blodgett, Abigail Fisher, Anthony Steed, Nikos Ntoumanis, Mark Hamer	
 Should an AI Assistant Change Its Appearance? Exploring Representation Strategies Across Everyday Contexts	Under review
<i>CHI Conference on Human Factors in Computing Systems (CHI 2027)</i>	
Jingyi Zhang, Anthony Steed	
 Perceived Agency During Agent-Avatar Control Transitions in Virtual Reality	Under review
<i>2027 IEEE Conference Virtual Reality and 3D User Interfaces (VR 2027)</i>	
Jingyi Zhang, Ruijun Sun, Klara Brandstätter, Nels Numan, Anthony Steed	
 Do You See What I See? Bring Live Pedestrians into an Outdoor Collaborative Mixed Reality Experience	Sep 2025
<i>38th ACM Symposium on User Interface Software and Technology (UIST 2025)</i>	
Jingyi Zhang, Ziwen Lu, Changrui Zhu, Simon Julier, Anthony Steed	
 Single Actor Controlling Multiple Avatars for Social Virtual Realities	Mar 2025
<i>2025 IEEE Conference Virtual Reality and 3D User Interfaces (VR 2025)</i>	
Jingyi Zhang, Anthony Steed	
 Supporting Co-Presence in Populated Virtual Environments by Actor Takeover of Animated Characters	Oct 2023
<i>2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2023)</i>	
Jingyi Zhang, Klara Brandstätter, Anthony Steed	
 Reviving the Euston Arch: A Mixed Reality Approach to Cultural Heritage Tours	Oct 2023
<i>2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct 2023)</i>	
Ziwen Lu, Jingyi Zhang , Kalila Shapiro, Nels Numan, Simon Julier, Anthony Steed	
 Comparing Mixed Reality Agent Representations: Studies in the Lab and in the Wild	Oct 2023
<i>29th ACM Symposium on Virtual Reality Software and Technology (VRST 2023)</i>	
Ben J Congdon, Gun Woo Park, Jingyi Zhang , Anthony Steed	
 Extending the Open Source Social Virtual Reality Ecosystem to the Browser in Ubiq	Oct 2023
<i>28th International ACM Conference on 3D Web Technology</i>	
Sebastian Friston, Ben Congdon, Nels Numan, Klara Brandstätter, Lisa Izzouzi, Felix Thiel, Jingyi Zhang , Daniele Giunchi, David Swapp, Anthony Steed	

Academic Service & Memberships

- Professional memberships: Member of ACM and IEEE.
- Peer review: Reviewer for ACM SIGCHI conferences, IEEE VR, and IEEE ISMAR.

Awards & Scholarship

Honorable Mention for Best Design Award ISMAR 2023 Student Competition	2023
 Rabin Ezra Scholarship	2023-2024

Skills

Programming Languages

Python, Java, C, C#, JavaScript, HTML, CSS, R, SQL, Haskell

Frameworks & Libraries

TensorFlow, PyTorch, scikit-learn, Hugging Face, OpenCV, NumPy, Pandas, Unity XR Interaction Toolkit, Meta SDK, OpenXR

Data & Analysis Environments

Jupyter Notebook, Google Colab, SPSS, MATLAB

Cloud & DevOps

Microsoft Azure, Docker, Git, GitHub

Other Interests :) (basic level)

Piano, Violin, Guzheng, Guqin, Guitar, Harmonica, and Marching Band (snare drum).

Operating Systems

Windows, macOS, Linux, iOS, Android

XR / VR Development

Unity, Meta Quest SDK, OpenXR Toolkit, Ubiq Networking Framework, Mixed Reality Toolkit (MRTK), Blender (basic)

Hardware & Prototyping

Arduino, Raspberry Pi, Processing

Design & Creative Tools

Figma, Adobe XD, Adobe Photoshop, Adobe Premiere Pro